

Artemis Agri Intelligence Platform

Hyperlocal climate monitoring, early warning alerts, and automated parametric insurance for smallholder farmers across Southern Africa.

Overview

Smallholder farmers in rural Namibia and Zimbabwe face devastating climate shocks — drought, flash floods, and frost — yet remain the least equipped to anticipate or recover from them. National weather forecasting is too broad to be actionable at farm level, leaving communities with no early warning and no time to take protective measures.

The **Artemis Agri Intelligence Platform** addresses this directly. We deploy networks of agricultural IoT sensors and hyperlocal weather monitoring stations within farming communities, feeding an intelligent alert engine that notifies farmers, community leaders, and insurers the moment conditions begin deteriorating. When predefined climate thresholds are breached, automated parametric insurance payouts are triggered — delivering financial relief within hours, not weeks.

Key Features

- **Hyperlocal weather monitoring** — ground-truth climate data at farm level
 - **Soil moisture sensing** — real-time soil condition tracking across multiple zones
 - **Automated early warning alerts** — SMS, app, and community broadcast notifications
 - **Valve control automation** — soil moisture data triggers direct irrigation responses
 - **Satellite monitoring** — landscape-scale crop and vegetation intelligence
 - **Parametric insurance triggers** — automated payout initiation on threshold breach
 - **AI-powered anomaly detection** — machine learning models trained on field data
 - **Real-time data dashboards** — live visualisation via Datacake
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Technology Stack

Layer	Technology
Sensors	Soil moisture sensors, weather stations
Connectivity	LoRaWAN, cellular (4G/LTE), satellite
Edge Processing	Node-RED, EG71 gateway
Data Protocol	MQTT
Cloud & Dashboards	Datacake
Satellite Intelligence	NDVI and vegetation index monitoring
AI & Analytics	Machine learning models for climate pattern detection
Alerts	SMS, mobile app, community broadcast

Pilot Results

We have completed three pilot deployments across three distinct regions in Namibia and Zimbabwe.

Current Active Pilot — Winter Crop Monitoring:

- 1 weather station
- 22 soil moisture sensors
- Automated valve control established
- Satellite monitoring across 420 hectares
- **99% data uptime** across all sensor nodes
- Positive farmer feedback across all three pilot cohorts

Infrastructure

The platform is built on Artemis Telecoms' existing connectivity infrastructure — operational LPWAN, cellular, and satellite networks already deployed across Namibia and Zimbabwe.

This ensures reliable data transmission in the remote farming environments where smallholder communities operate.

Impact Goal

30 new smallholder farms onboarded in Namibia within 12 months, with full early warning alert capability and parametric insurance integration established by end of programme.

About the Team

Bernard Chiwome — CEO & Project Lead Qualified veterinarian with 16 years of experience as a State Veterinarian in rural Namibia. Founder of AgriProject (farmer support consultancy) and Artemis Telecoms (IoT infrastructure and network operator).

Wesley Bruwer — Co-Founder & CTO Founder of IT Guru, a successful ISP. Architect behind Artemis Telecoms' network deployments across Namibia and Zimbabwe, with hands-on experience deploying LPWAN, cellular, and satellite connectivity across agricultural, municipal, and wildlife environments.

Contact

Artemis Telecoms Namibia & Zimbabwe